

Stages 2 and 3 of did_multiplegt_dyn

Where Stage 1's controls outputs are consumed: long-difference residualization and the $U^{\text{var},X}$ variance correction

Notes on L4582–L4954 of did_multiplegt_dyn.ado

Stage 1 (the previous note) produced, for each baseline level d ,

- $\hat{\theta}_d \in \mathbb{R}^K$ — `coefs_sq_‘d’_XX`;
- $\text{Den}_d^{-1} \in \mathbb{R}^{K \times K}$ — `inv_Denom_‘d’_XX`;
- the per-control score series $\text{prod}_X X_{k,g,t}$ — `prod_Xk_Ngt_XX`;
- the cell-mean fitted value $\hat{E}_{g,t}^{(d)}$ — `E_y_hat_gt_int_‘d’_XX`.

This note walks through how Stages 2 and 3 spend each of those four objects. **Stage 2** replaces the long-difference of Y with a controls-residualized version; **Stage 3** adds a variance term that accounts for the fact that $\hat{\theta}_d$ is estimated, not known. Effects-side only; placebos are symmetric and covered briefly at the end.

1 Notation refresh

$\ell \in \{1, \dots, L\}$	event-study horizon (‘i’ in the loop).
$d \in \{1, \dots, L_D\}$	baseline treatment level (<code>d_sq_int_XX</code>).
F_g	first switch date for group g (<code>F_g_XX</code>); $F_g = T + 1$ for never-switchers.
T_g	last period where g still has a control group (<code>T_g_XX</code>).
$S_g \in \{0, 1\}$	switcher direction: 1 = switcher-in ($\pm = +$), 0 = switcher-out ($\pm = -$).
$D_{g,1}$	baseline treatment of group g .
$Y_{g,t}, X_{k,g,t}$	outcome and k -th control.
$\Delta Y_{g,t}^{(\ell)} = Y_{g,t} - Y_{g,t-\ell}$	long-difference of Y , <code>diff_y_‘1’_XX</code> .
$\Delta X_{k,g,t}^{(\ell)} = X_{k,g,t} - X_{k,g,t-\ell}$	long-difference of X_k , <code>diff_Xk_‘1’_XX</code> .
$N_{g,t}$	weight of cell (g, t) (<code>N_gt_XX</code>).
G	number of groups (<code>G_XX</code>).
$N_\ell^\pm$	number of switcher- $\pm$ groups eligible at horizon ℓ (<code>N_‘+’_‘1’_XX</code>).

Throughout, $\pm$ stands for “+ for switchers-in (`increase_XX==1`), – for switchers-out (`increase_XX==0`)”. The code generates both families by setting `increase_XX` in an outer loop, so the same lines build + and – versions in turn.

2 Stage 2 — Long-difference residualization [ado L4582–4689]

Step 2.1 — Build the long differences of each control

For each horizon ℓ (looping over $\ell = 1, \dots, L$ outside this block) and each control k [ado L4586–4598]:

$$\Delta X_{k,g,t}^{(\ell)} = X_{k,g,t} - X_{k,g,t-\ell}, \quad (1)$$

$$\Delta X_{k,g,t}^{(\ell),N} = N_{g,t} \cdot \Delta X_{k,g,t}^{(\ell)}. \quad (2)$$

Stored as `diff_Xk_‘1’_XX` and `diff_Xk_‘1’_N_XX`.

Step 2.2 — The treatment-side moment $M_{d,\ell,k}^{\pm}$ [ado L4611–4622]

The companion-paper vectors $\mathbf{m}_{g,d,\ell}^{\pm}$ have one coordinate per control k . The code builds them in three passes.

Per- (g, t) summand [ado L4612]:

$$m_{g,d,\ell,k,t}^{\pm} = \mathbb{1}\{\ell \leq T_g - 2, D_{g,1} = d\} \cdot \frac{G}{N_{\ell}^{\pm}} \cdot \left[\text{dist}_{g,t}^{(\ell)} - \frac{N_{\ell,g}^{\pm,t}}{N_{g,t}^{c,(\ell)}} \cdot \text{nc}_{g,t}^{(\ell)} \right] \cdot \mathbb{1}\{\ell+1 \leq t \leq T_g\} \cdot \Delta X_{k,g,t}^{(\ell),N}. \quad (3)$$

The four named quantities (all built earlier in the file) are:

- $\text{dist}_{g,t}^{(\ell)}$ — distance-to-switch indicator (`distance_to_switch_‘1’_XX`): equals 1 when (g, t) is at relative time ℓ for switcher g .
- $\text{nc}_{g,t}^{(\ell)}$ — never-changer indicator (`never_change_d_‘1’_XX`): 1 if g is still untreated at t (control side).
- $N_{\ell,g}^{\pm,t}$ — cohort-size count (`N_+’_t_‘1’_g_XX`).
- $N_{g,t}^{c,(\ell)}$ — count of controls available for g at horizon ℓ (`N_gt_control_‘1’_XX`).

Sum across t within g [ado L4615]:

$$m_{g,d,\ell,k}^{\pm} = \sum_t m_{g,d,\ell,k,t}^{\pm}, \quad (4)$$

kept on the first row of each group ($\dots = .$ if `n != 1`, L4616).

Average over groups [ado L4620–4622]:

$$M_{d,\ell,k}^{\pm} = \frac{1}{G} \sum_g m_{g,d,\ell,k}^{\pm}. \quad (5)$$

Stored as `M_+’_d_‘k’_‘1’_XX`.

What this is. $M_{d,\ell,k}^{\pm}$ is the cross-group average of the k -th control’s long-difference, weighted by the residual of the distance-to-switch indicator on its control-group mean. It is the “treatment-side” summary that the variance correction multiplies $(\text{Den}_d^{-1} \cdot \text{score}_d - \hat{\theta}_d)$ by.

Step 2.3 — The cell-mean fitted value $\widehat{E}_{g,t}^{(d)}$ [ado L4625–4647]

The Stage-1 fitted value $\widehat{E}_{g,t,\text{int}}^{(d)} = \text{E_y_hat_gt_int_‘d’_XX}$ is now masked to zero in cells with too few control groups:

$$\widehat{E}_{g,t}^{(d,\ell)} = \widehat{E}_{g,t,\text{int}}^{(d)} \cdot \mathbb{I}\{E_{d,\ell}^{\text{denom}}(t) \geq 2\}, \quad (6)$$

where $E_{d,\ell}^{\text{denom}}(t)$ counts the available not-yet-switcher control groups (or clusters, if `cluster()` is used) at time t inside baseline d [ado L4625–4641]. Stored as `E\_y\_hat\_gt\_‘d’\_XX`.

Step 2.4 — The per-group control-side score $\text{in_sum}_{k,d}$ [ado L4653–4669]

Two scalar building blocks. First, the within-baseline window-size weight [ado L4657–4659]:

$$N_d^c = \sum_{(g,t)} N_{g,t} \cdot \mathbb{I}\{D_{g,1} = d, 2 \leq t \leq T_d, t < F_g, \Delta Y_{g,t} \neq \text{miss}\}, \quad (7)$$

where $T_d = \max_{h:D_{h,1}=d} F_h - 1$ [ado L4578–4580].

Second, the per- (g, t) score summand [ado L4664]:

$$\sigma_{k,g,t}^{(d)} = \frac{\text{prod_X}_{k,g,t} \cdot \delta_{g,t}^{(d)} \cdot (\Delta Y_{g,t} - \widehat{E}_{g,t}^{(d,\ell)}) \cdot \mathbb{I}\{2 \leq t \leq F_g - 1\}}{N_d^c}, \quad (8)$$

with the DoF adjustment

$$\delta_{g,t}^{(d)} = 1 + \mathbb{I}\{E_{d,\ell}^{\text{denom}}(t) \geq 2\} \cdot \left[\sqrt{\frac{E_{d,\ell}^{\text{denom}}(t)}{E_{d,\ell}^{\text{denom}}(t) - 1}} - 1 \right] \quad (9)$$

[ado L4661–4663]. The adjustment is the standard small-sample correction when demeaning by an estimated cell mean.

Then collapse to one observation per group [ado L4669]:

$$\boxed{\text{in_sum}_{k,d}(g) = \sum_t \sigma_{k,g,t}^{(d)}} \quad (10)$$

Stored as `in\_sum\_‘k’\_‘d’\_XX`.

What this is. $\text{in_sum}_{k,d}(g)$ is group g ’s contribution to the score of the auxiliary regression of ΔY on ΔX (with FEs) for baseline cell d , after double-residualization. It is the “control-side” summary fed into Den_d^{-1} in Stage 3.

Step 2.5 — The actual residualization of $\Delta Y^{(\ell)}$ [ado L4679]

$$\boxed{\Delta Y_{g,t}^{(\ell)} \leftarrow \Delta Y_{g,t}^{(\ell)} - \sum_{k=1}^K \widehat{\theta}_d[k] \cdot \Delta X_{k,g,t}^{(\ell)} \text{ if } D_{g,1} = d.} \quad (11)$$

In the code this is written as a loop over k that subtracts one product at a time. Effectively $\Delta Y_{g,t}^{(\ell)}$ is overwritten in place with the controls-residualized long difference. After this line, the rest of the event-study machinery (the part not specific to controls) treats `diff\_y\_‘1’\_XX` as if it were the raw outcome long-difference – controls have vanished from the surface.

Important nuance [ado L4680]. The replacement is unconditional on $\Delta X_{k,g,t}^{(\ell)}$ being non-missing. If any control is missing, $\Delta Y_{g,t}^{(\ell)}$ becomes missing and the cell is silently excluded from the long-difference estimation. This is by design: it prevents the estimator from using (g, t) cells where the residualization is undefined.

3 Stage 3 — The $U^{\text{var},X}$ correction [ado L4905–4954]

After Stage 2, the per-group variance score with no controls is, from L4904,

$$U_{G,g,\ell}^{(\pm, \text{noX})}(t) = \mathbb{K}\{\ell \leq T_g - 1\} \cdot \frac{G}{N_\ell^\pm} \cdot \left[\text{dist}_{g,t}^{(\ell)} - \frac{N_{\ell,g}^{\pm,t}}{N_{g,t}^{c,(\ell)}} \text{nc}_{g,t}^{(\ell)} \right] \cdot \mathbb{K}\{\ell+1 \leq t \leq T_g\} \cdot N_{g,t} \cdot \text{DOF}_{g,t}^{(\ell)} \cdot (\Delta Y_{g,t}^{(\ell)} - \widehat{E}_{g,t}^{(\ell)}). \quad (12)$$

With controls, Stage 3 adds a second term coming from the fact that $\widehat{\theta}_d$ is estimated. The block at L4908–4954 builds that second term, `part2_switch‘+’_‘1’_XX`, group by group.

Step 3.1 — The bracketed score $\text{br}_{d,j}$ [ado L4921–4938]

For each baseline d and each control coordinate j , the inner loop over k [ado L4924–4935] computes

$$[\text{Den}_d^{-1} \cdot \text{in_sum}_d]_j(g) = \sum_{k=1}^K \text{Den}_d^{-1}[j, k] \cdot \text{in_sum}_{k,d}(g), \quad (13)$$

restricted to $\mathbb{K}\{D_{g,1} = d, F_g \geq 3\}$ [ado L4929]. Then $\widehat{\theta}_d[j]$ is subtracted [ado L4938]:

$$\boxed{\text{br}_{d,j}(g) = [\text{Den}_d^{-1} \cdot \text{in_sum}_d]_j(g) - \widehat{\theta}_d[j].} \quad (14)$$

Stored as `in.brackets_‘d’_‘j’_XX`.

What this is. The asymptotic linearization of $\widehat{\theta}_d - \theta_d$ on the control sample is $\widehat{\theta}_d - \theta_d \approx \text{Den}_d^{-1} \cdot \text{in_sum}_d / G + o_p(\cdot)$. So $\text{br}_{d,j}(g)$ is, up to sign, the influence-function coordinate of $\widehat{\theta}_d - \theta_d$ attributable to group g .

Step 3.2 — The per-group correction $\text{part2}_\ell^\pm$ [ado L4941–4950]

Multiply the influence function by the treatment-side moment and sum across controls [ado L4942–4945]:

$$\text{combined}^{\pm, d, \ell}(g) = \sum_{j=1}^K M_{d,\ell,j}^\pm \cdot \text{br}_{d,j}(g), \quad (15)$$

then sum over baseline cells [ado L4950]:

$$\boxed{\text{part2}_\ell^\pm(g) = \sum_d \text{combined}^{\pm, d, \ell}(g).} \quad (16)$$

Stored as `part2_switch‘+’_‘1’_XX`.

Step 3.3 — Per-group variance score $U_{G,g,\ell}^{\pm,\text{var},X}$ [ado L4957 +]

Sum the no-controls piece over t within g [ado L4957] and add the controls correction:

$$U_{G,g,\ell}^{\pm,\text{var},X} = \underbrace{\sum_t U_{G,g,\ell}^{(\pm,\text{noX})}(t)}_{\text{U_Gg'1'_{var_XX}}} + \text{part2}_{\ell}^{\pm}(g). \quad (17)$$

Step 3.4 — Asymptotic variance of the effect estimator

Once $U_{G,g,\ell}^{\pm,\text{var},X}$ is built for every group g , the event-study variance of effect ℓ is the standard cross-group second moment:

$$\widehat{\text{Var}}(\widehat{\text{Effect}}_{\ell}^{\pm}) = \frac{1}{G^2} \sum_{g=1}^G (U_{G,g,\ell}^{\pm,\text{var},X})^2 \xrightarrow{\text{cluster}(c)} \frac{1}{G^2} \sum_c \left(\sum_{g \in c} U_{G,g,\ell}^{\pm,\text{var},X} \right)^2, \quad (18)$$

matching the cluster-robust convention implemented elsewhere in the file. The $\text{part2}_{\ell}^{\pm}(g)$ term is what makes (??) the *right* variance when controls are in the model; without it, the SEs would understate the noise arising from $\hat{\theta}_d$ estimation.

4 Visualizing the dependency graph

Where each Stage 1 output enters Stages 2–3:

$\hat{\theta}_d$	→ Stage 2 , eq. (??), the residualization of $\Delta Y^{(\ell)}$.
	→ Stage 3 , eq. (??), subtracted from the influence function.
Den_d^{-1}	→ Stage 3 , eq. (??), scaling of the score.
$\text{prod_X}_{k,g,t}$	→ Stage 2 , eq. (??), the score summand.
$\hat{E}_{g,t}^{(d)}$	→ Stage 2 , eq. (??)/(??), the demean inside the score.

5 Cross-reference table — Stage 2

Math symbol	.ado variable	Defined at
$\Delta X_{k,g,t}^{(\ell)}$	diff_Xk_‘1’_XX	L4593
$\Delta X_{k,g,t}^{(\ell),N}$	diff_Xk_‘1’_N_XX	L4598
$m_{g,d,\ell,k,t}^{\pm}$ (per- t)	m_‘+’_g_‘k’_‘d’_‘1’_XX	L4612
$m_{g,d,\ell,k}^{\pm}$ (sum over t)	m_‘+’_‘d’_‘k’_‘1’_XX	L4615
$M_{d,\ell,k}^{\pm}$ (avg over g)	M_‘+’_‘d’_‘k’_‘1’_XX	L4621
$E_{d,\ell}^{\text{denom}}(t)$	E_hat_denom_‘k’_‘d’_XX	L4635/L4640
$\widehat{E}_{g,t}^{(d,\ell)}$	E_y_hat_gt_‘d’_XX	L4647
$T_d = \max_{h:D_{h,1}=d} F_h - 1$	T_d_XX	L4579
N_d^c	N_c_‘d’_XX	L4659
$\delta_{g,t}^{(d)}$ (DoF adjustment)	in_sum_temp_adj_‘k’_‘d’_XX	L4662–4663
$\sigma_{k,g,t}^{(d)}$ (score summand)	in_sum_temp_‘k’_‘d’_XX	L4664
$\text{in_sum}_{k,d}(g)$	in_sum_‘k’_‘d’_XX	L4669
residualization $\Delta Y^{(\ell)} - = \theta_d^{(k)} \Delta X_k^{(\ell)}$	(overwrites diff_y_‘1’_XX)	L4679

6 Cross-reference table — Stage 3

Math symbol	.ado variable	Defined at
$[\text{Den}_d^{-1} \cdot \text{in_sum}_d]_{j,k}$ (term)	in_brackets_‘d’_‘j’_‘k’_temp_XX	L4929
$[\text{Den}_d^{-1} \cdot \text{in_sum}_d]_j$ (sum over k)	in_brackets_‘d’_‘j’_XX (incremented)	L4934
$\text{br}_{d,j}(g)$ (subtract $\widehat{\theta}_d[j]$)	in_brackets_‘d’_‘j’_XX (final)	L4938
$M_{d,\ell,j}^{\pm} \cdot \text{br}_{d,j}(g)$	combined_‘+’_temp_‘d’_‘j’_‘1’_XX	L4942
$\text{combined}^{\pm,d,\ell}(g)$ (sum over j)	combined_‘+’_temp_‘d’_‘1’_XX	L4945
$\text{part2}_{\ell}^{\pm}(g)$ (sum over d)	part2_switch_‘+’_‘1’_XX	L4950
$U_{G,g,\ell}^{(\pm,\text{noX})}(t)$ summand	U_Gg_‘1’_temp_var_XX	L4904
$U_{G,g,\ell}^{\pm,\text{var},X}$	U_Gg_‘1’_var_XX (+ part2_switch_‘+’_‘1’_XX)	L4957 + ...

7 Placebos — the symmetric story [ado L5145+]

The placebo path repeats Stages 2 and 3 with three substitutions:

- **Long differences are flipped in time.** [ado L5156]

$$\Delta X_{k,g,t}^{(\ell),\text{pl}} = X_{k,g,t-\ell} - X_{k,g,t-2\ell},$$

so the placebo lives entirely *before* the switch date.

- **$\widehat{\theta}_d$ is reused.** [ado L5177] The same `coefs_sq_‘d’_XX` computed in Stage 1 multiplies $\Delta X_{k,g,t}^{(\ell),\text{pl}}$ to residualize `diff_y_pl_‘1’_XX`. There is no separate “placebo θ_d ”. This is by design: under parallel trends and no-anticipation, θ_d is identified pre-switch, exactly where placebos live.
- **Den_d^{-1} and $\text{in_sum}_{k,d}$ are reused.** The placebo variance correction `part2_pl_switch_‘+’_‘1’_XX` multiplies the same influence function $\text{br}_{d,j}$ by a placebo treatment-side moment $M_{d,\ell,j}^{\pm,\text{pl}}$ [ado L5165–5172].

The bookkeeping renames (`m_pl`, `M_pl`, `in_sum_pl`, `part2_pl`) but the math is identical to (??)–(??) with the placebo $\Delta X^{(\ell), \text{pl}}$ replacing the effect $\Delta X^{(\ell)}$.

8 Why this design is elegant

Three properties make this implementation robust:

1. **Stage 1 is computed once.** $\hat{\theta}_d$, Den_d^{-1} , $\text{prod_X}_{k,g,t}$, $\hat{E}_{g,t,\text{int}}^{(d)}$ depend only on (g, t) data, not on the horizon ℓ . They are recycled across every ℓ and across effects vs. placebos with no additional cost.
2. **Stage 2 is just an additive shift.** The residualization at L4679 overwrites $\Delta Y^{(\ell)}$ in place. After it, every downstream piece of code that has nothing to do with controls consumes the residualized long-difference as if it were the raw one. Adding HDFE (the previous note) requires no change to that downstream code.
3. **Stage 3 is a clean additive correction.** The $U^{\text{var}, X}$ term (??) is added to the no-controls variance score in one line at L4957+. Switching off controls (`'controls'=="`) skips L4908–4954 entirely and recovers the no-controls variance unchanged.

The cost of this elegance is two nested loops over K^2 at L4921–4946, which is fine for typical $K \leq 5$ and dominates only when the user passes many controls. If you ever add HDFE-derived controls (e.g. thousands of dummies via `absorb()`), keep the absorbed dimensions out of K – that is exactly what the HDFE plan does.